

PHYS 225

Week 2, Lecture 2:
The Three Fundamental Effects of Relativity



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reminders

- Homework 1 due at 11:59 pm this Thursday, 1/29
- Please, “tag” each question of your HW in Gradescope with the appropriate pages!
- Office Hours **tomorrow 1/28 at 4:00 - 7:00 pm, Loomis 276**
Thursday 1/29 at 5:00 - 7:00 pm, Loomis 279

1. Introduction to Special Relativity;

2. Fundamental Effects of Relativity;

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3. Vectors, Matrices, and Index Notation;
 4. Derivation of Lorentz Transformations;
 5. Matrix Multiplication & Taylor Series;
 6. Symmetries and Group Theory;
 7. Invariant Interval & Spacetime Metric;
 8. Relativistic Energy, Momentum, and Force; Separation of Variables in 1D;
 9. Relativistic Force in 2D; Vector Fields, Curvilinear Coordinates, Vector Differential Operators;
 10. 4-vector Manipulation, Doppler Effect;
 11. Relativistic Kinematics;
 12. Line and Surface Integrals, Stokes' Theorem, Differential Forms of Maxwell's Equations;
 13. Complex Numbers and Fourier Transforms;

- GPS
- Aviation
- High-Speed Spacecrafts and Space Research
- High-Speed Particle Physics

Week 2, Lecture 2

The Three Fundamental Effects

Today we will:

- Carefully define the concepts of reference frame, event, simultaneity, time, and length
- Use these concepts to see the the three fundamental effects of relativity:
 - Rear Clock Ahead
 - Time Dilation
 - Lorentz (length) Contraction

One Minute Paper: List one or two examples of systems that you know from PHYS 211 that can be used as clocks. In other words, their motion is cyclical with a definite period.

Week 2, Lecture 2

The Three Fundamental Effects

From last time: “What if you’re traveling at the speed of light, and someone else shines a laser in the same direction as you — what speed do you see the laser traveling at?”

$$v_{\text{rel}} = \frac{c - c}{1 - c^2/c^2} = \frac{0}{0}!$$

A reference frame traveling at the speed of light is **not a valid inertial frame**. Rather, c is the **upper bound** on the speed of inertial frames.

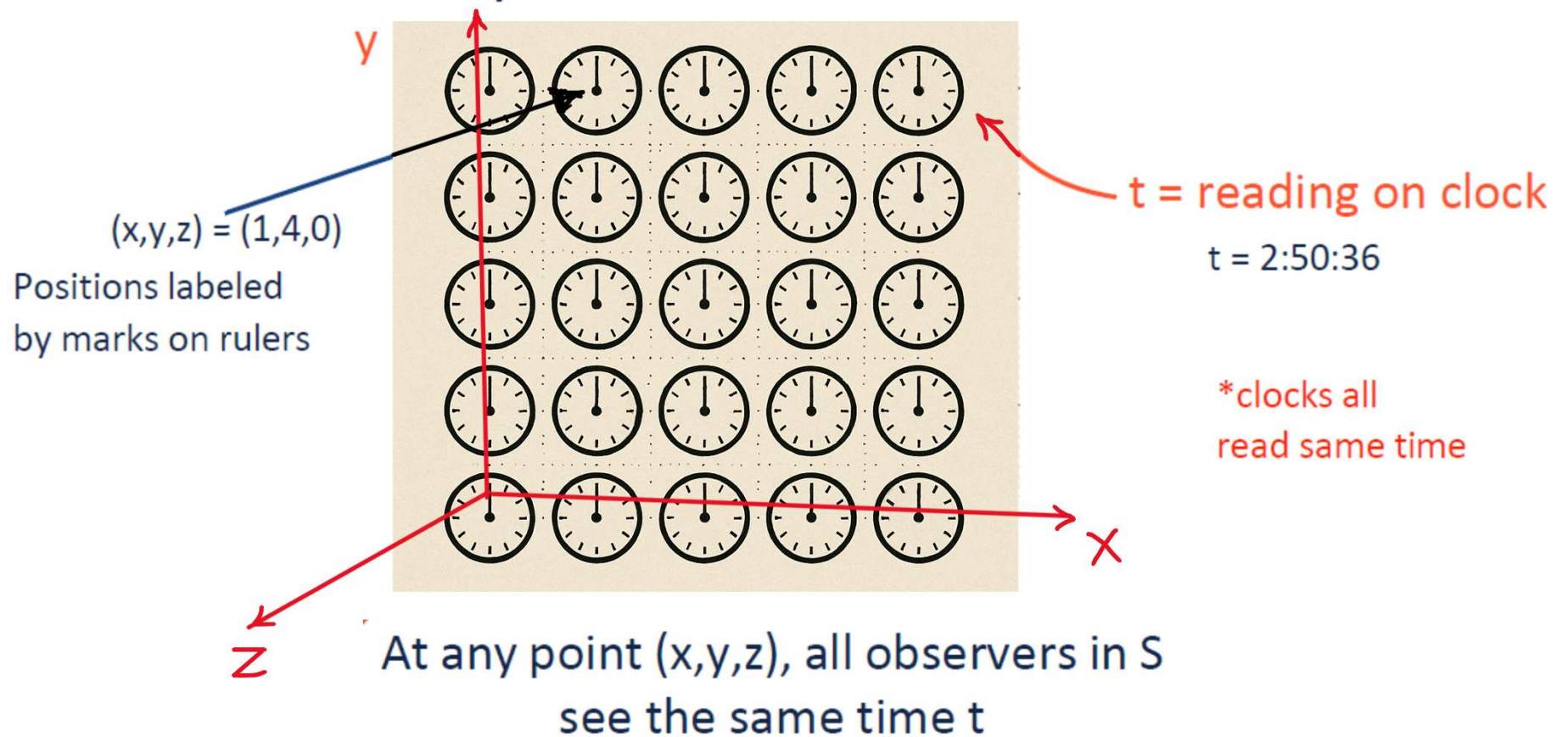
For any $v < c$, you will find $v_{\text{rel}} = c$ (as you showed in discussion)

Week 2, Lecture 2

The Three Fundamental Effects

What is a reference frame (call it S)?

Network of synchronized* clocks and rulers



Week 2, Lecture 2

The Three Fundamental Effects

What is a ruler?

Rigid rod of fixed
length in S

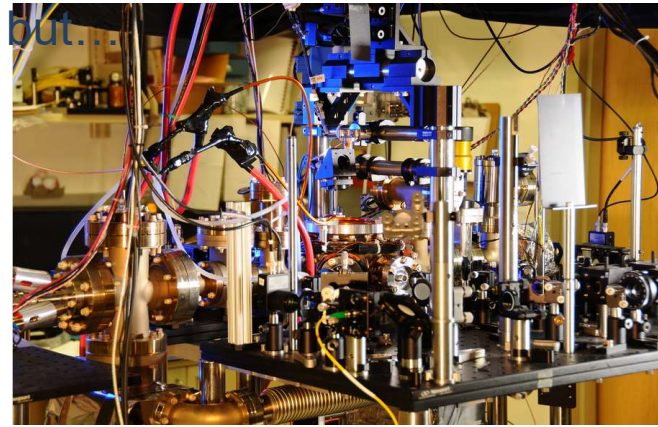


Until 1960, the SI standard of length was disseminated using platinum-iridium meter bars such as these from the NIST Museum.

What is a clock?

Any periodic system

Any pendulum can be a clock,
but...



<https://www.nist.gov/news-events/news/2013/08/nist-ytterbium-atomic-clocks-set-record-stability>

The second is currently defined **using cesium atoms, which absorb and emit microwave radiation with a specific frequency**. Atomic clocks count 9,192,631,770 of those microwave oscillations, and we call the elapsed time interval a second.

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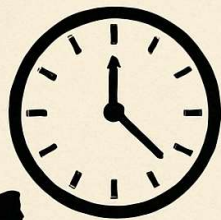
The Three Fundamental Effects

What is an event?

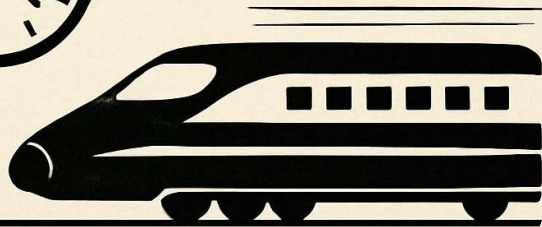
Something that happens at a **specific time and place**

SPECIAL RELATIVITY

7:00 PM April 24



$$E=mc^2$$



(x,y) = Green and Goodwin

z = second floor

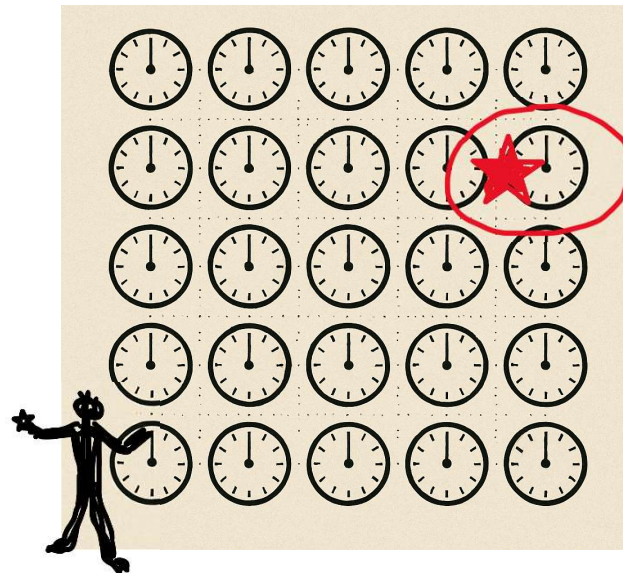
t = 4:00 PM as seen on a
clock in Loomis 151

Week 2, Lecture 2

The Three Fundamental Effects

What is the time of an event in S ?

The reading on the clock in S at the location of the event



(this may seem pedantic, don't all clocks in S have the same reading?
Yes, but this will not be true from the perspective of another observer:
remember the light-on-a-train example from discussion!)

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The Three Fundamental Effects

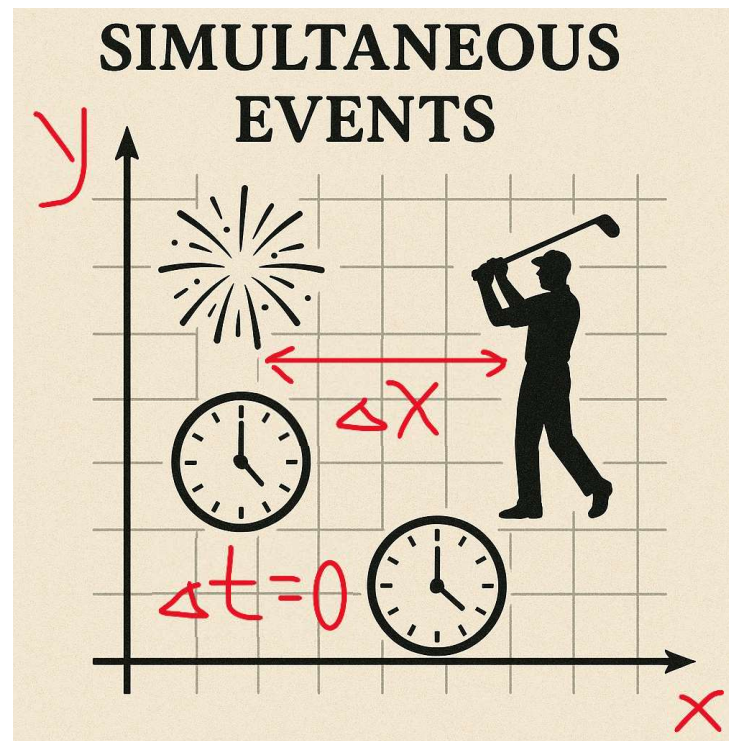
Two events are simultaneous in S
if they occur at **different locations** in S ,
and their respective clocks read the **same time**

Bang

$x = 2$ a.u.

$y = 6$ a.u.

$t = 4.5:00$ PM



Swing

$x = 7$ a.u.

$y = 5$ a.u.

$t = 4.5:00$ PM

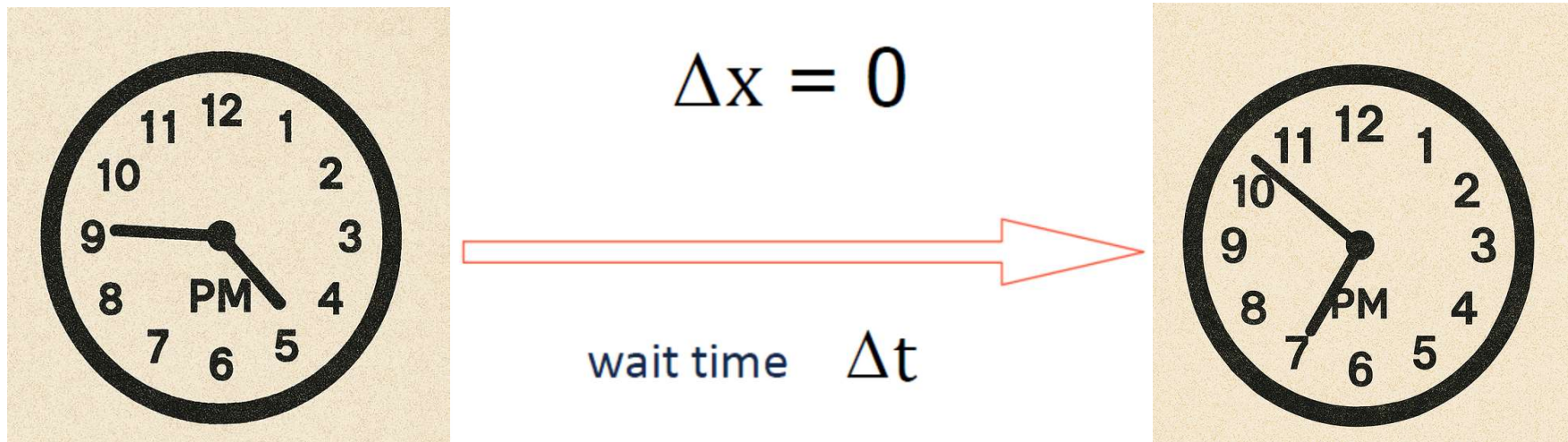
a.u. – arbitrary units

Week 2, Lecture 2

The Three Fundamental Effects

How do we measure **time elapsed in S**?

Look at the **same** clock (i.e. at the same position), at two **different** times, and take the difference of the readings

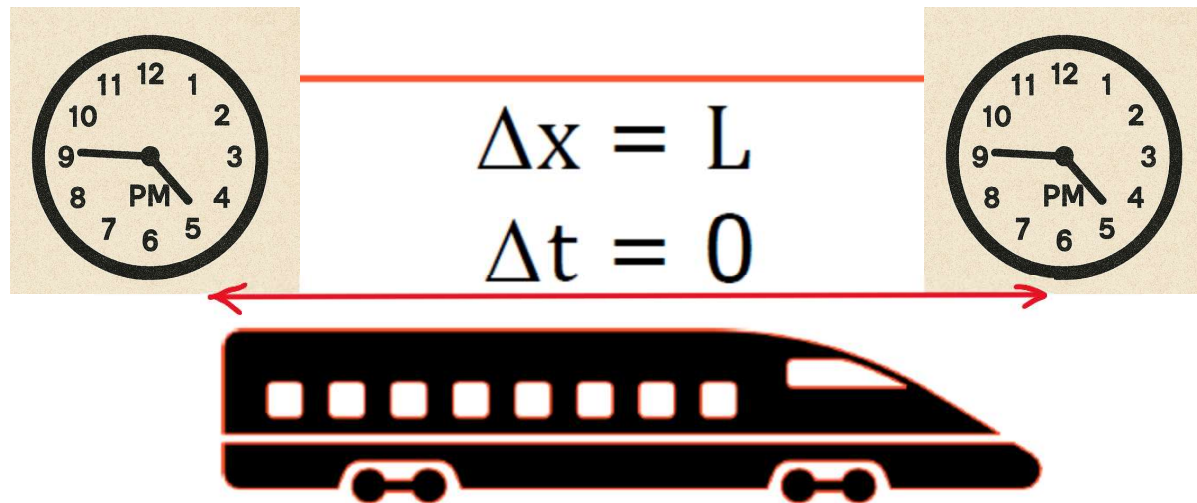


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The Three Fundamental Effects

How do we measure a **length in S**?

It's the **distance** between the two ends of an object, measured **simultaneously** (i.e. at the same time in S)

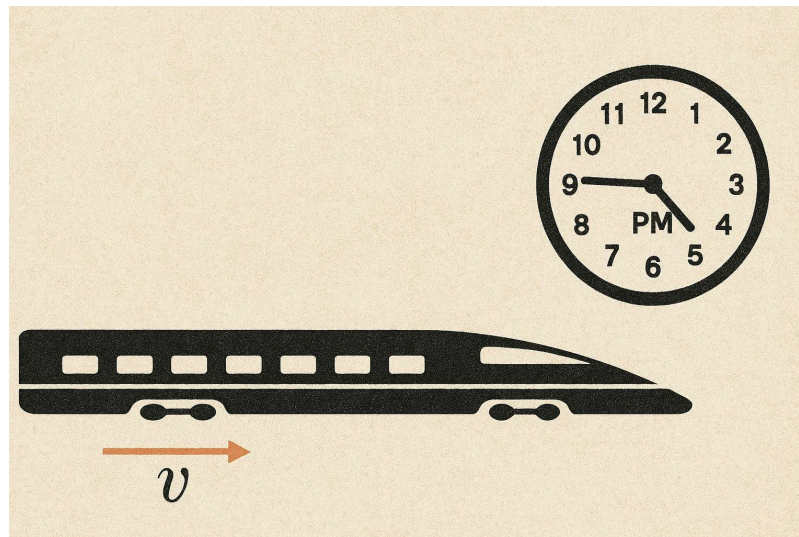


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The Three Fundamental Effects

How do we measure a **length**
(of an object at constant velocity) in S ?

Equivalently, it's the time elapsed on a **single clock** in S while the object passes in front of the clock



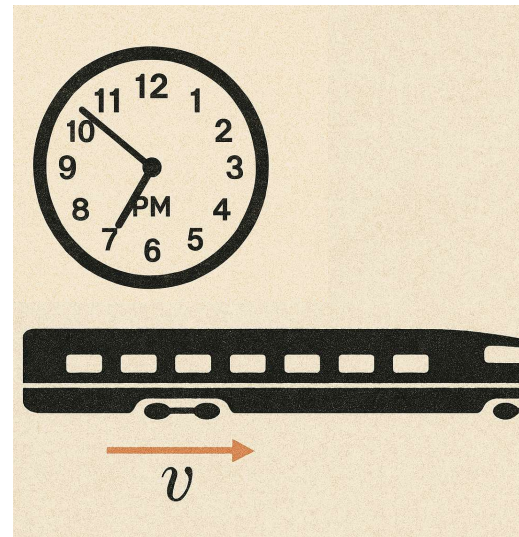
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The Three Fundamental Effects

How do we measure a **length**
(of an object at constant velocity) in S?

Equivalently, it's the time elapsed on a **single clock** in S while the object passes in front of the clock

$$L_{\text{obs}} = v\Delta t$$



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The Three Fundamental Effects

In all of what follows:

Frame S' is moving at speed v with respect to S



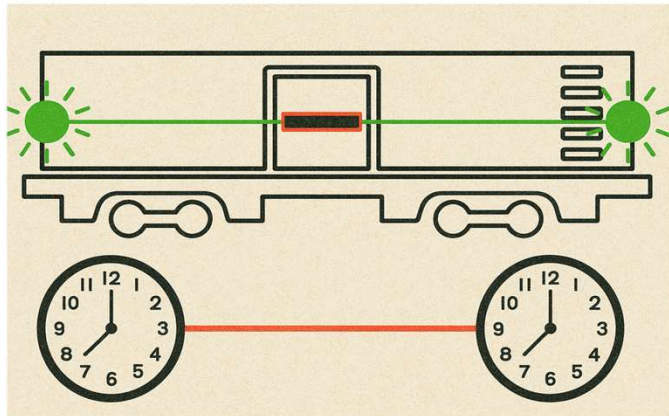
Week 2, Lecture 2

The Three Fundamental Effects

Fundamental Effect 1: Rear Clock Ahead

Think back to last week's discussion:

In S' , if we put two lasers exactly at the center, light hitting front and back of train is simultaneous:



Two events are simultaneous in S' if they occur at **different locations** in S' , and their respective clocks read the **same time**

Week 2, Lecture 2

The Three Fundamental Effects

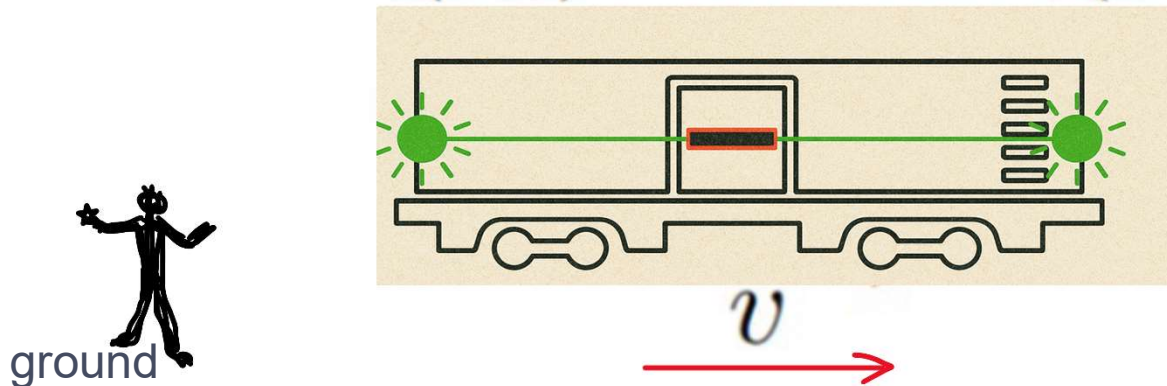
Fundamental Effect 1: Rear Clock Ahead

Think back to last week's discussion:

In S, light hits front and back of train at **different times**

$$t_B = \frac{L_S}{2(c + v)}$$

$$t_F = \frac{L_S}{2(c - v)}$$



Note: L_S is the length of the train **as measured in S**

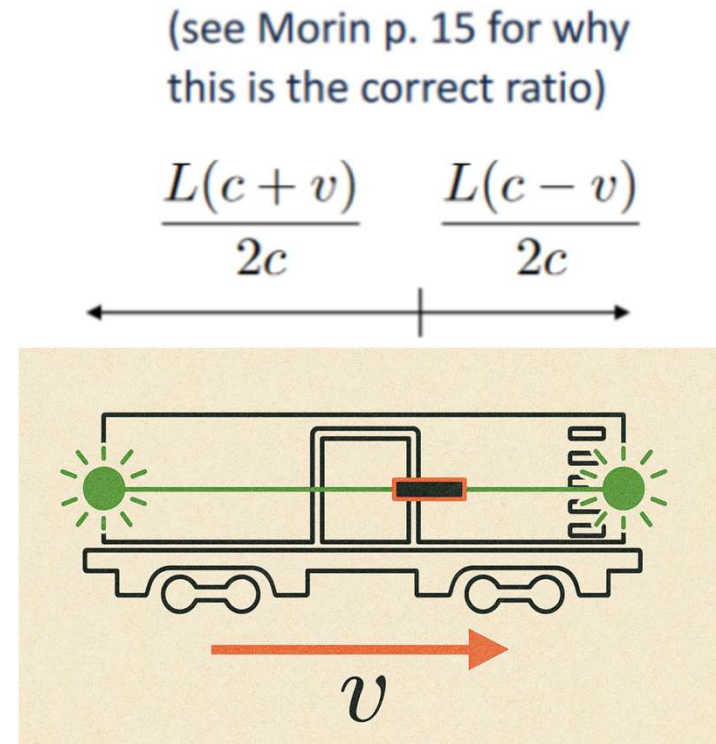
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The Three Fundamental Effects

Fundamental Effect 1: Rear Clock Ahead

Let's ask a slightly different question:
if the light hits the front and back of
the train **simultaneously in S**,
what do the train clocks read?

Note: L is the length of the train **as measured in S'**



Week 2, Lecture 2

The Three Fundamental Effects

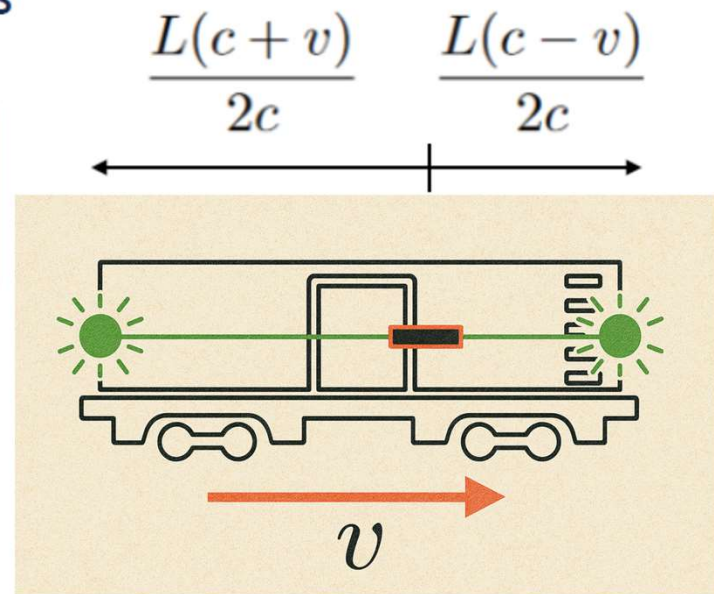
Fundamental Effect 1: Rear Clock Ahead

In the train frame S' , the front-moving light has a “head start”. Light takes longer to travel to the back, so if the front train clock reads zero, the rear train clock reads

$$\frac{L(c+v)}{2c^2} - \frac{L(c-v)}{2c^2} = \frac{Lv}{c^2}$$

This is a statement about the readings on two different S' clocks, at the same S time.

Note: L is the length of the train as measured in S'



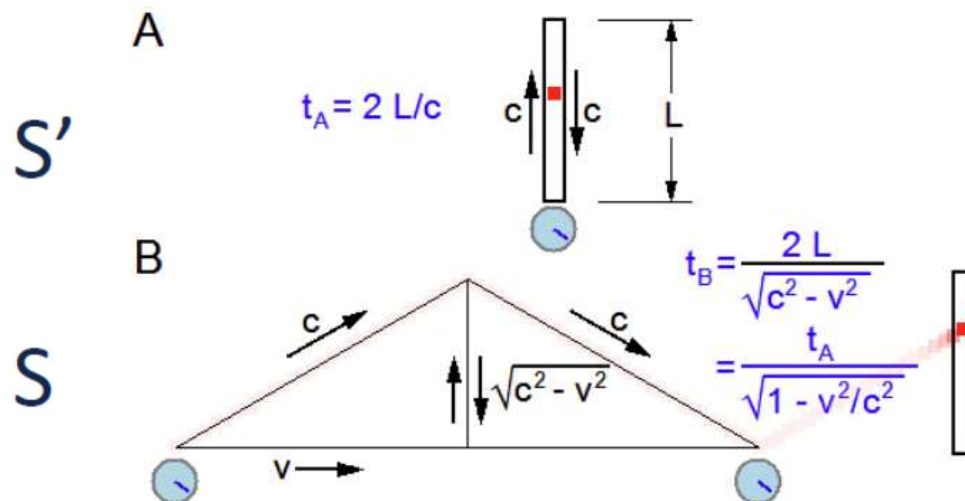
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The Three Fundamental Effects

Fundamental Effect 2: Time Dilation

How do we measure **time elapsed in S'** ?

Look at the **same S' clock** at two **different** times



In S , the vertical component of the light's velocity is reduced! (Discussion 2)

More time passes for you:
time in S' appears dilated

Time dilation is a statement about the rate (or difference in readings) of a **single clock**

Week 2, Lecture 2

The Three Fundamental Effects

Gamma Factors and Proper Time

The diagram shows the equation $\Delta t = \gamma \Delta \tau$ enclosed in a red rounded rectangle. Arrows point from descriptive text to the symbols in the equation: Δt is labeled 'Time elapsed in S', $\Delta \tau$ is labeled 'Proper time (time elapsed in clock's rest frame, S\')', and γ is labeled 'Gamma factor (Lorentz factor, boost factor)'. The gamma factor is further defined as $\frac{1}{\sqrt{1 - v^2/c^2}}$.

$$\Delta t = \gamma \Delta \tau$$

Time elapsed in S

Proper time (time elapsed in clock's rest frame, S')

Gamma factor (Lorentz factor, boost factor)

$$\frac{1}{\sqrt{1 - v^2/c^2}}$$

You'll derive this in Discussion 2

Week 2, Lecture 2

The Three Fundamental Effects

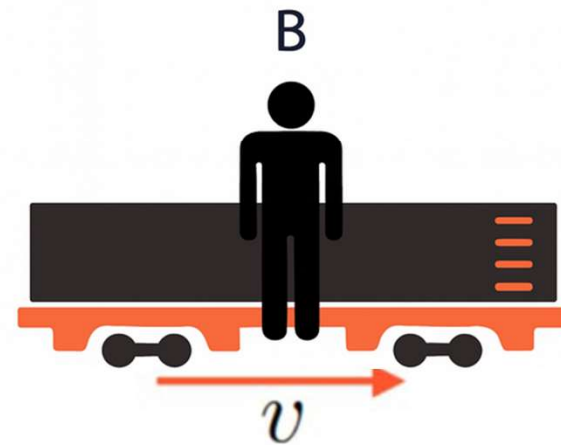
The Twin Paradox

$$\gamma \geq 1 :$$

less time elapses for a moving observer than a stationary one



"I'm stationary, and B is moving,
so B's clocks run slower."

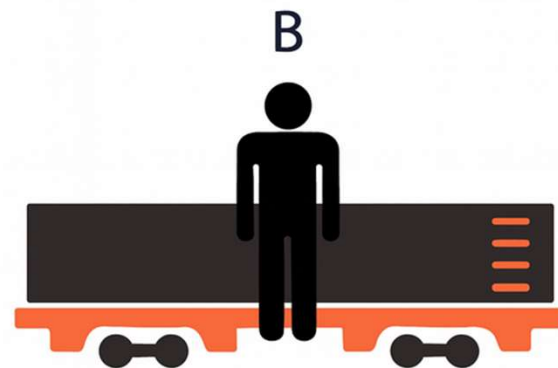


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The Three Fundamental Effects

The Twin Paradox

$\gamma \geq 1$: less time elapses for a moving observer than a stationary one



"I'm stationary, and A is moving, so A's clocks run slower."

Week 2, Lecture 2

The Three Fundamental Effects

The Twin Paradox

$\gamma \geq 1$: less time elapses for a moving observer than a stationary one

Who's right?
They both are!

A is comparing
two S clocks to
a single S' clock

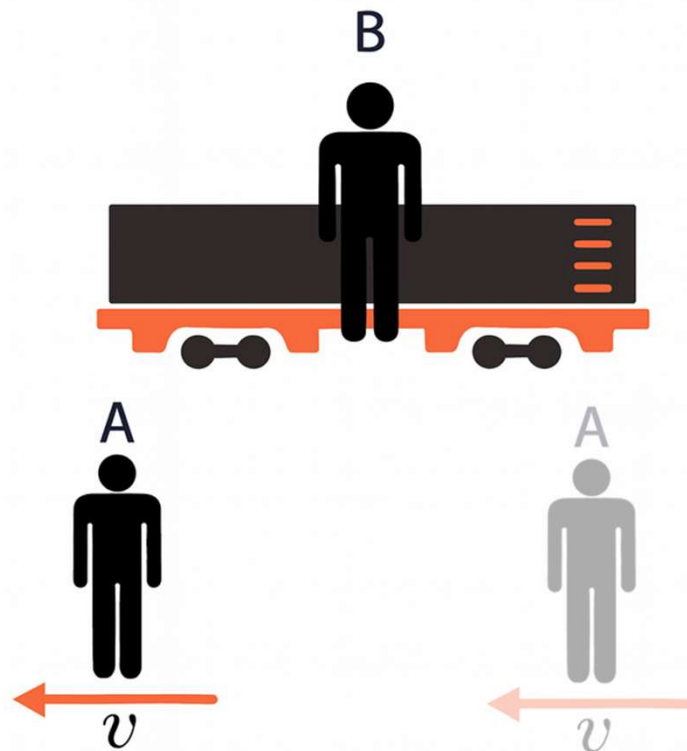
If B turns around to
compare his clock with A,
he has to accelerate:
he's not an inertial observer!
B will return younger than A

B is comparing
two S' clocks
to a single S clock

Week 2, Lecture 2

The Three Fundamental Effects

Fundamental Effect 3: Length Contraction



B measures the length of his train to be L , in the rest frame of the train

$$\Delta t' = \frac{L}{v}$$

Less time passes on A's watch:

$$\Delta t = \frac{L}{\gamma v}$$

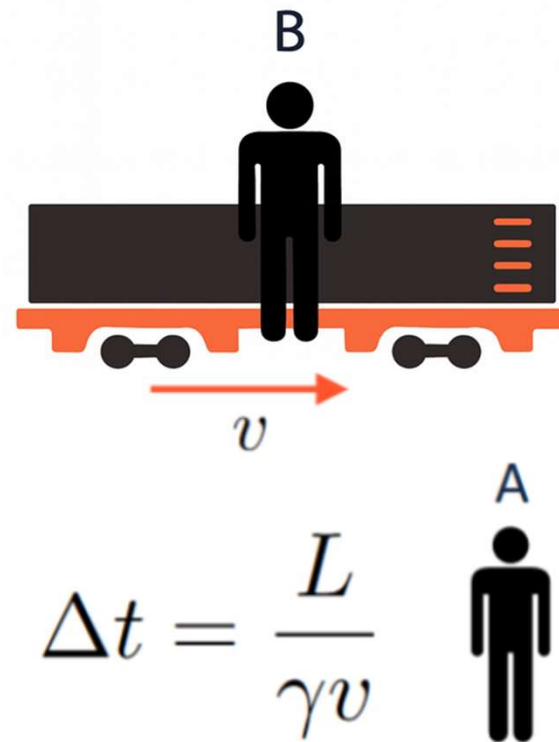
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The Three Fundamental Effects

Fundamental Effect 3: Length Contraction

How do we measure
a **length** in **S**?

Time elapsed on a **single
clock** in **S** while the
object passes in front of
the clock



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The Three Fundamental Effects

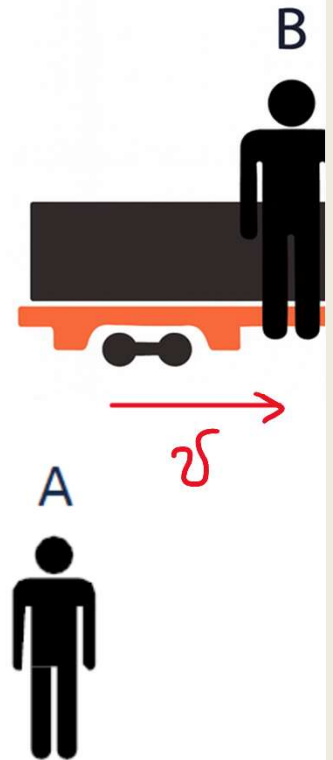
Fundamental Effect 3: Length Contraction

How do we measure
a **length** in **S**?

Time elapsed on a **single
clock** in **S** while the
object passes in front of
the clock

$$L_A = v \Delta t = \frac{L}{\gamma}$$

$$\Delta t = \frac{L}{\gamma v}$$



Week 2, Lecture 2

The Three Fundamental Effects

Fundamental Effect 3: Length Contraction

Proper length (as measured
in object rest frame)

$$L_{\text{obs}} = \frac{L_{\text{proper}}}{\gamma}$$

Length as
measured in S

Moving objects are shorter.

Measuring proper length requires simultaneous events.
Measuring observed length requires simultaneous events or comparing two clocks.

Week 2, Lecture 2

The Three Fundamental Effects

Things we can all agree on:
the coordinates of a single event,
as measured in both S and S' .

A (in S) hits B's clock (in S') with a hammer as B (with velocity v) passes A.
This breaks the clock, so its reading can be compared with A's later.

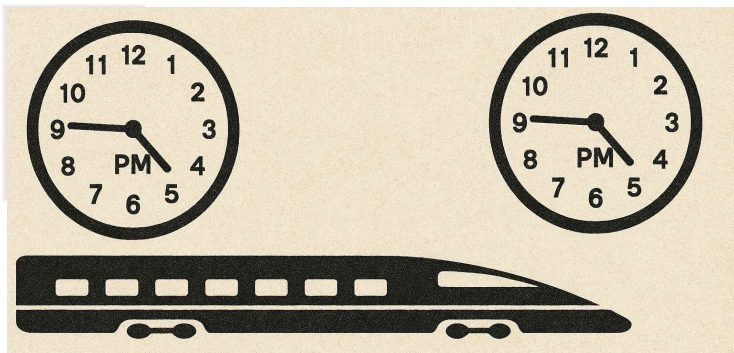
In less flowery language: **A hits B's clock with a hammer**
is an event that takes place at (t, x) in S and (t', x') in S' . We have to agree on those numbers to be able to define frames in the first place!

Week 2, Lecture 2

The Three Fundamental Effects

Summary

$$+ \frac{L_{\text{proper}} v}{c^2}$$



Rear clock ahead

$$\Delta t = \gamma \Delta \tau$$

Time dilation

$$L_{\text{obs}} = \frac{L_{\text{proper}}}{\gamma}$$

Lorentz (length) contraction

One Minute Paper: which of the fundamental effects was the most surprising?